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DS210 A1
3/26/25
Collaborators: none

My first function takes in a file name and translates it into a hashmap by making a key separating key and value by ":" and a vector of strings for the value by separating everything after ":" by ",", trimming for spaces, making everything lowercase, and filtering for empty values. This function returns a Result enum because opening the file may result in an error.

My second function "name_likes_recipe" takes in a name and recipe and outputs a result enum of bool or string. True if the name and recipe are in the hash maps and the recipe contains at least 60% of the categories liked by the name, false if the name and recipe are in the hash maps and the recipe contains less than 60% of the categories liked by the name. It returns a string of Err if either the name or recipe does not exist in the hash maps. The function also takes in three hash maps to use for comparisons. I reversed the categories and ingredients hash map so that each ingredient was a key and the value was a category. This made creating the hash set of categories for the recipe far easier.

My third function, "liked_recipes," takes in a name and the same hash maps as before. This function returns a result enum as well. Ok() of a vector of strings that contains all the recipes that a name likes or Err() of string if the name does not exist. This function runs through all the recipes and checks if a name is like the recipe using the "name_likes_recipe" function. It stores all the liked values in a vector of strings to return later.

My fourth function, "popular_recipes," takes in only hashmaps and returns a tuple of the top three most liked recipes and their like counts. This function runs through all the recipes and then calculates how many people like the recipe by looping over all the names for each recipe and updating the top three variables. I address the equal values issue by using a comparison operator and swaps, but this method will not work properly unless all recipes have equal lengths. I did not account for the differing lengths of numbers.

In the main function, I create the different hashmaps from the files, create the reversed hashmaps, and take in the user input. I make the input lowercase to make it case insensitive. My hashmaps are also lowercase to ensure case insensitivity. I choose which function to run using an if, else if, and else statement. If the input contains a "," that means that it is a name and recipe running the name_likes_recipe function. If the input is "popular recipe", it runs the popular_recipe function. In any other case, it runs the liked_recipes function.

I also have tests that check most cases. I check all the cases for name_likes_recipe, if liked_recipes works, and if popular_recipes works, by creating a smaller data set and calculating values by hand.

AI use:

I used ChatGPT to help me create the “file_to_hashmap” function. I asked Chat how to make a vector of values from a line of words separated by “,”. It told me that to use “.split()”, “.map()”, “.into_iter()”, and “.collect()”. I split the values by “,”, mapped it after trimming so that there was no spaces, and made it lowercase. I used into_iter() to make the vector an iterator and then .collect() to put all the values into the vector of strings.

I also asked it how to filter out empty strings and it told me to use “.filter(|s: &String| !s.is_empty())”. This code checks every value and checks if the value is empty or not.

```
// file to hashmap converter
fn file_to_hashmap(filename:&str) -> io::Result<HashMap<String,Vec<String>>> {
    let file: File = File::open(filename)?;
    let reader: io::BufReader<File> = io::BufReader::new(file);

    let mut map: HashMap<String, Vec<String>> = HashMap::new();

    for line in reader.lines(){
        let line: String = line?;

        if let Some((key,values)) = line.split_once(":") {
            let value_list: Vec<String> = values.split(",").map(|s:
&str|s.trim().to_lowercase()).filter(|s: &String|
!s.is_empty()).into_iter().collect();
            map.insert(key.trim().to_lowercase(), value_list);
        }
    }
    return Ok(map)
}
```

Output for user entry:

Highlight: input

Bold: output

(base) riwazshrestha@crc-dot1x-nat-10-239-100-59 p1 % cargo run

Compiling p1 v0.1.0 (/Users/riwazshrestha/Desktop/DS210/Homework/HW6/p1)

Finished `dev` profile [unoptimized + debuginfo] target(s) in 0.32s

Running `target/debug/p1`

Enter Command (name), (popular recipes), (name,recipe):

halle vaughn

halle vaughn likes these recipes: Ok(["recipe819", "recipe892", "recipe575", "recipe616", "recipe198", "recipe258", "recipe721", "recipe582", "recipe591", "recipe485", "recipe278", "recipe188", "recipe216", "recipe661", "recipe875", "recipe735", "recipe294", "recipe509", "recipe879", "recipe637", "recipe202", "recipe719", "recipe451", "recipe601", "recipe752", "recipe839", "recipe997", "recipe553", "recipe444", "recipe972", "recipe175", "recipe988", "recipe812", "recipe712", "recipe952", "recipe403", "recipe636", "recipe811", "recipe760", "recipe158", "recipe609", "recipe415", "recipe920", "recipe300", "recipe893", "recipe166", "recipe260", "recipe694", "recipe426", "recipe124", "recipe617", "recipe889", "recipe103", "recipe134", "recipe653", "recipe317", "recipe393", "recipe463", "recipe488", "recipe364", "recipe140", "recipe361", "recipe418", "recipe37", "recipe660", "recipe196", "recipe706", "recipe357", "recipe555", "recipe686", "recipe758", "recipe738", "recipe390", "recipe702", "recipe883", "recipe718", "recipe953", "recipe500", "recipe740", "recipe532", "recipe228", "recipe90", "recipe749", "recipe161", "recipe86", "recipe168", "recipe773", "recipe48", "recipe457", "recipe853", "recipe257", "recipe899", "recipe938", "recipe165", "recipe957", "recipe651", "recipe761", "recipe701", "recipe392", "recipe367", "recipe273", "recipe157", "recipe424", "recipe22", "recipe731", "recipe388", "recipe212", "recipe113", "recipe765", "recipe925", "recipe708", "recipe471", "recipe845", "recipe832", "recipe320", "recipe476", "recipe977", "recipe330", "recipe658", "recipe477", "recipe111", "recipe125", "recipe525", "recipe857", "recipe827", "recipe971", "recipe396", "recipe299", "recipe671", "recipe778", "recipe484", "recipe184", "recipe570", "recipe675", "recipe145", "recipe115", "recipe683", "recipe47", "recipe430", "recipe122", "recipe856", "recipe830", "recipe588", "recipe186", "recipe454", "recipe468", "recipe104", "recipe798", "recipe62", "recipe870", "recipe265", "recipe814", "recipe345", "recipe922", "recipe379", "recipe46", "recipe21", "recipe322", "recipe756", "recipe681", "recipe696", "recipe6", "recipe178", "recipe74", "recipe822", "recipe266", "recipe949", "recipe284", "recipe829", "recipe219", "recipe991", "recipe634", "recipe492", "recipe81", "recipe356", "recipe647", "recipe440", "recipe101", "recipe215", "recipe666", "recipe442", "recipe707", "recipe199", "recipe927", "recipe385", "recipe253", "recipe291", "recipe935", "recipe406", "recipe503", "recipe340", "recipe542", "recipe854", "recipe801", "recipe552", "recipe646", "recipe517", "recipe755", "recipe233", "recipe520", "recipe352", "recipe963", "recipe187", "recipe737", "recipe355", "recipe823", "recipe15", "recipe154", "recipe325", "recipe469", "recipe625", "recipe474", "recipe848", "recipe906", "recipe171", "recipe490", "recipe421", "recipe143", "recipe413", "recipe213", "recipe533", "recipe254", "recipe560", "recipe100", "recipe804", "recipe763", "recipe716", "recipe290",

"recipe224", "recipe809", "recipe211", "recipe557", "recipe495", "recipe826", "recipe506",
"recipe497", "recipe750", "recipe382", "recipe429", "recipe473", "recipe423", "recipe796",
"recipe897", "recipe572", "recipe589", "recipe640", "recipe370", "recipe252", "recipe437",
"recipe493", "recipe959", "recipe547", "recipe621", "recipe551", "recipe301", "recipe272",
"recipe44", "recipe127", "recipe564", "recipe207", "recipe629", "recipe255", "recipe412",
"recipe507", "recipe742", "recipe342", "recipe821", "recipe475", "recipe136", "recipe39",
"recipe491", "recipe614", "recipe156", "recipe75", "recipe571", "recipe231", "recipe985",
"recipe374", "recipe865", "recipe974", "recipe907", "recipe45", "recipe95", "recipe757",
"recipe744", "recipe793", "recipe59", "recipe332", "recipe36", "recipe725", "recipe909",
"recipe787", "recipe928", "recipe180", "recipe334", "recipe680", "recipe397", "recipe979",
"recipe579", "recipe654", "recipe656", "recipe174", "recipe71", "recipe544", "recipe970",
"recipe639", "recipe446", "recipe846", "recipe162", "recipe241", "recipe860", "recipe841",
"recipe900", "recipe574", "recipe29", "recipe297", "recipe3", "recipe494", "recipe862",
"recipe905", "recipe26", "recipe422", "recipe8", "recipe478", "recipe102", "recipe998",
"recipe5", "recipe436", "recipe595", "recipe63", "recipe72", "recipe508", "recipe610",
"recipe353", "recipe445", "recipe833", "recipe221", "recipe960", "recipe344", "recipe176",
"recipe566", "recipe324", "recipe886", "recipe943", "recipe501", "recipe27", "recipe642",
"recipe863", "recipe428", "recipe612", "recipe286", "recipe76", "recipe583", "recipe987",
"recipe714", "recipe409", "recipe19", "recipe794", "recipe665", "recipe784", "recipe995",
"recipe259", "recipe528", "recipe908", "recipe337", "recipe859", "recipe691", "recipe606",
"recipe282", "recipe596", "recipe505", "recipe950", "recipe842", "recipe308", "recipe238",
"recipe363", "recipe611", "recipe815", "recipe435", "recipe523", "recipe962", "recipe51",
"recipe976", "recipe416", "recipe902", "recipe771", "recipe855", "recipe285", "recipe736",
"recipe619", "recipe279", "recipe269", "recipe69", "recipe541", "recipe25", "recipe343",
"recipe177", "recipe346", "recipe4", "recipe954", "recipe384", "recipe670", "recipe577",
"recipe785", "recipe131", "recipe149", "recipe152", "recipe918", "recipe235", "recipe462",
"recipe759", "recipe214", "recipe398", "recipe695", "recipe271", "recipe657"])

(base) riwazshrestha@crc-dot1x-nat-10-239-100-59 p1 % cargo run

Finished `dev` profile [unoptimized + debuginfo] target(s) in 0.00s

Running `target/debug/p1`

Enter Command (name), (popular recipes), (name,recipe):

Halle Vaughn, recipe5

halle vaughn likes recipe5: Ok(true)

(base) riwazshrestha@crc-dot1x-nat-10-239-100-59 p1 % cargo run

Finished `dev` profile [unoptimized + debuginfo] target(s) in 0.00s

Running `target/debug/p1`

Enter Command (name), (popular recipes), (name,recipe):

popular recipes

The most liked recipes are: ("recipe533", 424, "recipe308", 403, "recipe621", 401)

(base) riwazshrestha@crc-dot1x-nat-10-239-100-59 p1 %